Python Analyzers

Set - A

Branch: _____ **Date:** 24-01-2026

Total Marks: 50 **Time:** 2.5 Hours

Sr. No.	GR ID	Name	Signature
1			

Rules & Guidelines

- Participants must bring their own laptop
- Use of the internet is strictly restricted during the competition (apart from the final submission)
- Any form of plagiarism or code sharing will lead to disqualification
- Participants must strictly follow the instructions given during the event
- Late submissions will not be accepted
- Maintain discipline during the competition
- The decision of the faculty/judges will be final

Submission Details

- **File format:** .ipynb file
- **File name format:** Name_GRID_PythonAnalyzers.ipynb
- **Submission method:** Submit the **.ipynb file** and the given **dataset (.csv/.xlsx)** and upload it to the GitHub repository. **Submit that repo link** in the **Google form** for submission.
- Submit here: **LINK**

🛠️ Mandatory Technologies

Students **must use only**:

- numpy
- pandas
- matplotlib
- seaborn

❌ Any other ML / AI libraries are **not allowed**

📂 Dataset Overview (Gigantic Real-World Dataset)

Dataset Name:
[Indian_Retail_Business_Sales_Gigantic_Dataset.csv]

Dataset Size:

- ~50,000 Rows
- 10+ Business Attributes

Dataset Description

This dataset represents **multi-year retail sales transactions across Indian cities and regions**, suitable for **business case study analysis**.

Columns Explanation

Column Name	Description
Order_ID	Unique order identifier
Order_Date	Date of transaction
Region	Sales region (North, South, etc.)
City	City where the sale occurred
Customer_Type	New or Returning customer
Product_Category	Product category
Quantity	Units sold

Unit_Price	Price per unit
Discount_Percent	Discount applied
Revenue	Final revenue after discount

🎯 Project Objective

Analyze the provided **retail business dataset** to:

- Clean and prepare real-world data
- Perform exploratory and business-driven analysis
- Identify trends, patterns, and performance metrics
- Visualize insights using charts
- Present meaningful business conclusions

📝 Mandatory Tasks (Exam Tasks for Students)

🔹 Task 1: Notebook Structure & Data Loading

- Proper title & introduction
- Import all required libraries
- Load the dataset correctly
- Display:
    - Dataset shape
    - First & last 5 rows
- Proper Markdown usage

🔹 Task 2: Data Cleaning & EDA

Students must:

- Check and handle missing values
- Validate data types
- Detect duplicates (if any)
- Generate:
    - Summary statistics

    - Category distributions
- Comment on initial observations

🔹 Task 3: Business Data Analysis

Perform a **minimum of 5 analytical tasks**, such as:

- Region-wise total revenue
- Category-wise sales performance
- Customer type comparison (New vs Returning)
- Monthly/yearly sales trends
- Impact of discounts on revenue

🔹 Task 4: Data Visualization

Create a **minimum of 5 meaningful visualizations**, such as:

- Bar chart (Category vs Revenue)
- Line chart (Time-based sales)
- Histogram (Revenue distribution)
- Pie plot (Proportion of Categories)
- Scatter Plots
- Subplots (Combine multiple plots)

Each chart must include:

- Title
- Axis labels
- Insight explanation

🔹 Task 5: Final Insights & Task Completion

- Summarize key findings
- Mention business insights
- Provide recommendations

📊 Evaluation & Marking Scheme (50 Marks)

Criteria	Marks
Notebook Structure & Data Loading	10
Data Cleaning & EDA	10
Data Analysis	10
Data Visualization	10
% Task Completion & Overall Quality	10
TOTAL	**50**

⚠️ Mandatory Rules

- Only .ipynb file accepted
- No plagiarism or copied notebooks
- Code must be readable & logical
- Insights are more important than quantity
- The Internet is allowed only for the final submission

Remember to follow the instructions provided professionally, make suitable assumptions wherever necessary, and avoid copying code or content from unauthorized sources.

Good luck with your project work!

Python Analyzers
Data Science & AI/ML Department

BRING ON YOUR CODING ATTITUDE